Two speeds at the boundary: zeros of sums of conjugate sections of power series

Kentaro Amauchi*

August 18, 2026

Abstract

Let $w_0, w_1, \ldots$ be non-negative reals, $G(z) = \sum_{j \geq 0} w_j z^j$ and $G_k(z) = \sum_{j<k} w_j z^j$ its sections. We study the zeros of

$$F_k(q) \;=\; 1 + G_k(2q) + G_k(2-2q),$$

a sum of two sections under substitutions that are complex conjugates of one another on the line $\operatorname{Re} q = \frac{1}{2}$, and in particular whether they approach the symmetry point $q = \frac{1}{2}$ as $k \to \infty$.

Jentzsch's theorem describes the zeros of a *single* section, and adding functions can cancel zeros; we found no result covering sums of sections. The observation this note is built on is that on the symmetry line F_k is *real-valued*, so its zeros there are sign changes — and a sign change cannot be cancelled. Two results follow. First, for constant weights we obtain a closed form $F_k(\frac{1}{2} + it) = A + w\rho^k \sin(k\theta)/t$ with $A = 1 + 2w_0 - 2w$, from which the zero nearest the symmetry point exists and satisfies $t_1 = \pi/2k\,(1 + O(1/k))$, and is given exactly by $t = \frac{1}{2}\tan(n\pi/k)$ when $A = 0$. Second, when the weights decay the rate changes: for $w_j \asymp (j+1)^{-s}$ we find $t_1 \asymp \sqrt{s \log k / 2k}$, slower by a factor $\asymp \sqrt{k \log k}$. The dividing line is the decay of w_j and not the convergence of $\sum_j w_j$ — a distinction that our own data did not test until a profile lying on the line was computed. The constant at $s = 1$ is reported as unsettled, with the two candidate laws we have refuted or failed to resolve.

1 The question

Throughout, $w_0, w_1, w_2, \ldots$ are non-negative real numbers, not all zero,

$$G(z) = \sum_{j \geq 0} w_j z^j, \qquad G_k(z) = \sum_{j=0}^{k-1} w_j z^j,$$

and $R \in (0, \infty]$ denotes the radius of convergence of G. The object of study is

$$F_k(q) \;=\; 1 + G_k(2q) + G_k(2-2q), \tag{1}$$

a polynomial of degree at most $k-1$, invariant under $q \mapsto 1-q$. We ask where its zeros go as $k \to \infty$, and specifically whether they approach the symmetry point $q = \frac{1}{2}$.

*Independent researcher. amkn.sub03@gmail.com, https://mathlab0911.github.io/, ORCID https://orcid.org/0009-0000-0890-4395

Why this is not immediate. The zeros of a *single* section are classical. **Jentzsch's theorem** (1914) states that if $0 < R < \infty$ then every point of $|z| = R$ is a limit point of the zeros of the sections G_k; Szegő's refinement gives angular equidistribution along a subsequence, and the result has been extended to Dirichlet, Kapteyn and Neumann series, to analytic curves and to ultrametric fields. Applied to G with $R = 1$, it says the zeros of G_k accumulate at $z = 1$, which is $q = \frac{1}{2}$ under $z = 2q$.

But (1) is a *sum* of two sections under affine substitutions, plus a constant, and adding functions can cancel zeros: nothing in the single-section theory survives the sum automatically. We searched for a result covering sums of sections and found none; §10 says exactly where we looked, so that a reader who knows otherwise can correct us cheaply.

Where the object comes from, and why it does not matter here. F_k arose elsewhere, as a one-parameter deformation of a quantity attached to a finite set of integers, in which the w_j are a normalised gap statistic. **Nothing below uses that**, and the note is written so that it *cannot*: every statement from here on is about w_j, G and F_k as defined above, and nothing else is assumed, cited or required.

1.1 Terminology

This note carries no vocabulary of its own, and there is therefore no terminology table. Everything below is stated in the standard language of power series: coefficients, radius of convergence, sections, and zeros. The absence of a table is the point rather than an omission — a reader should be able to check every statement here without acquiring a single term.

2 The reduction: on the symmetry line the object is real

2.1 Why these two substitutions

A reader may reasonably ask why (1) pairs $G_k(2q)$ with $G_k(2-2q)$ rather than with some other companion. The answer is that this is the only affine pairing for which the two arguments are complex conjugates along a line, which is what makes §2 available.

Concretely, let $\varphi_1(q) = \alpha q+\beta$ and $\varphi_2(q) = \gamma q+\delta$ with real coefficients. Then $\varphi_2(q) = \overline{\varphi_1(q)}$ for all q on a vertical line $\operatorname{Re} q = c$ exactly when $\gamma = -\alpha$ and $\delta = \beta + 2\alpha c$ — that is, when φ_2 is φ_1 reflected in that line. Taking $\alpha = 2$, $\beta = 0$, $c = \frac{1}{2}$ gives $\varphi_2(q) = 2 - 2q$, which is (1).

Remark 2.1 (what the pairing costs and buys). [proved; elementary.] The reflection is what creates the difficulty and also what resolves it. It creates it because $G_k(\varphi_1) + G_k(\varphi_2)$ is a genuine sum, to which the single-section theory does not apply. It resolves it because on the mirror line the sum is $2 \operatorname{Re} G_k(\varphi_1)$, a real function — and the zeros of a real function of one variable are visible by the intermediate value theorem, with no complex analysis at all.

This is the note in one paragraph. The rest is the work of finding, for each class of weights, a place where that real function is negative.

Proposition 2.2 (reduction)**.** *[proved. Two independent derivations along disjoint routes (r203, r205); verified numerically against direct summation at 80 digits, worst agreement 50 digits,* `trackm_r203`*.] Let the w_j be real. Then for every real t,*

$$F_k\left(\tfrac{1}{2} + it\right) \;=\; 1 + 2 \operatorname{Re} G_k(1 + 2it) \;=\; 1 + 2 \sum_{j<k} w_j \, \rho^j \cos(j\theta),$$

where $\rho = |1+2it| = \sqrt{1 + 4t^2}$ and $\theta = \arg(1+2it) = \arctan 2t$. In particular F_k is real-valued on $\operatorname{Re} q = \frac{1}{2}$, and every sign change of $t \mapsto F_k(\frac{1}{2} + it)$ is a zero of F_k on that line.

Proof. With $q = \frac{1}{2} + it$ we have $2q = 1 + 2it$ and $2 - 2q = 1 - 2it = \overline{2q}$. Since the w_j are real, $G_k(\bar{z}) = \overline{G_k(z)}$, so $G_k(2q) + G_k(2 - 2q) = 2\,\mathrm{Re}\,G_k(1 + 2it)$. Writing $1 + 2it = \rho e^{i\theta}$ gives the sum. □ □

Two remarks, and the second is the whole method of this note.

Remark 2.3 (the shape of the sum). [immediate from Proposition 2.2.] The displayed sum is a cosine series with *growing* amplitudes: $\rho > 1$ for $t \neq 0$, so the coefficients $w_j \rho^j$ may increase even when the w_j decrease. Every estimate below is a competition between that growth and that decay.

Remark 2.4 (the cancellation problem disappears on the line). [proved; it is the content of Proposition 2.2.] The obstruction to transporting Jentzsch is that two sections may cancel each other's zeros. On $\mathrm{Re}\,q = \frac{1}{2}$ the object is a real function of one real variable, and **a sign change of a real function cannot be cancelled by anything**. So the on-line zeros are exactly the sign changes, and their existence is a question about a real function — which is why the results below are elementary where the general problem is not.

Remark 2.5 (what the reduction does *not* say). [measured; the counterexample is explicit, `pinch_r202`.] Proposition 2.2 says the line is where F_k is real. **It does not say the zero nearest $q = \frac{1}{2}$ lies on it.** For $w_j = (c/2)^j$ with $c < 2$ — so $R > 1$ — it does not: at $c = 1$, $k = 70$ the nearest *on-line* zero is at distance 0.872 while the nearest zero is at 0.504, certified by the argument principle. The nearest zero lies on the line in the boundary regime $R = 1$, where it was confirmed by an argument-principle count for $r \leq 0.25$, and this note asserts it only there.

We record the point because we imported the on-line assumption across the two regimes ourselves, silently, and a winding-number certificate is what caught it.

3 Constant weights: a closed form, and the pinch

Theorem 3.1 (constant weights). *[proved. Written proof below; independently re-derived along a disjoint route by a second reader who did not see it (r205); consequences* (a)–(e) *and the three constants of §3.1 re-derived in that reading. Numerically verified,* `trackm_r203`*.] Let $w_0 \geq 0$ and $w_j = w > 0$ for $1 \leq j \leq k-1$, and put*

$$A := 1 + 2w_0 - 2w, \qquad \rho = \sqrt{1 + 4t^2}, \qquad \theta = \arctan 2t.$$

Then for every $t > 0$,

$$F_k\left(\tfrac{1}{2} + it\right) = A + w\,\rho^k \frac{\sin(k\theta)}{t}, \tag{2}$$

exactly. Consequently:

(a) *$F_k(\frac{1}{2} + it) \to 1 + 2w_0 + 2(k-1)w$ as $t \to 0^+$, the value at the symmetry point;*

(b) *if $A \geq 0$ then $F_k(\frac{1}{2} + it) > 0$ for $0 < \theta < \pi/k$: no zero before the first rung;*

(c) *at $\theta = 3\pi/2k$, that is $t = \frac{1}{2}\tan(3\pi/2k)$, one has $\sin(k\theta) = -1$ and hence $F_k \leq A - w\,\frac{4k}{3\pi}(1 + o(1)) < 0$ for all large k;*

(d) *therefore F_k has a zero on the line with*

$$\tfrac{1}{2}\tan(\pi/k) < t_1 \leq \tfrac{1}{2}\tan(3\pi/2k), \qquad \textit{so} \qquad t_1 = \frac{\pi}{2k}(1 + O(1/k));$$

(e) *if $A = 0$ the zero set on the line is* exactly $t = \frac{1}{2}\tan(n\pi/k)$, $n = 1, 2, \ldots, \lfloor (k-1)/2 \rfloor$.

Proof. The sum is geometric: $G_k(z) = w_0 + w \sum_{j=1}^{k-1} z^j = w_0 + w\,(z^k - z)/(z-1)$, and on the line $z - 1 = 2it$, so

$$\mathrm{Re}\,\frac{z^k - z}{2it} = \frac{\mathrm{Im}(z^k - z)}{2t} = \frac{\rho^k \sin(k\theta) - 2t}{2t} = \frac{\rho^k \sin(k\theta)}{2t} - 1.$$

Substituting into Proposition 2.2 gives $F_k = 1 + 2w_0 + 2w\big[\rho^k \sin(k\theta)/2t - 1\big]$, which is (2).

For (a), $\theta = 2t + O(t^3)$ and $\rho^k \to 1$ as $t \to 0$, so $\sin(k\theta)/t \to 2k$ and $F_k \to A + 2kw = 1 + 2w_0 + 2(k-1)w$. For (b), $\sin(k\theta) > 0$ on $0 < \theta < \pi/k$ while $w, \rho^k, t > 0$, so $F_k > A \geq 0$. For (c), $t = \frac{1}{2}\tan(3\pi/2k) \leq (3\pi/4k)(1 + O(k^{-2}))$ and $\rho^k \geq 1$, so $w\rho^k/t \geq 4kw/3\pi\,(1 + o(1))$, which exceeds the constant A once $k > 3\pi A/4w$. Then (b) and (c) give opposite signs at the two ends of $[\frac{1}{2}\tan(\pi/k), \frac{1}{2}\tan(3\pi/2k)]$ and F_k is continuous there, so it vanishes inside; both endpoints are $\frac{\pi}{2k}(1 + O(1/k))$, which is (d). For (e), $A = 0$ leaves $w\rho^k \sin(k\theta)/t$, and w, ρ^k, t are non-zero for $t > 0$. □ □

3.1 Three instances, and why one is special

The families that motivated this note are all instances of Theorem 3.1, distinguished only by the value of A.

weights	w_0	w	A	zero set on the line
$w_0 = 0,\ w_j = \frac{1}{2}$	0	$\frac{1}{2}$	**0**	*exactly* $t = \frac{1}{2}\tan(n\pi/k)$
$w_0 = 1,\ w_j = \frac{1}{2}$	1	$\frac{1}{2}$	2	bracketed, $t_1 = \frac{\pi}{2k}(1 + O(1/k))$
$w_0 = 1,\ w_j = 1$	1	1	1	the same

Remark 3.2 (one identity, three constants). [proved, being Theorem 3.1 evaluated three times.] These three had been measured, cited and counted separately as evidence before the general case was written down. They are one identity with one constant, and the row admitting a closed form is exactly the row with $A = 0$, i.e. the row whose w_0 vanishes.

Separately computed instances of one mechanism look like independent evidence and are not: the count was a count of computations, not of cases. Stating the general member also *explains* which row is special, which no amount of further measurement would have done.

3.2 The $A = 0$ family, written out

The row with $A = 0$ is worth seeing in full, because it is the one case in this note where the answer is a formula rather than a bracket. Take $w_0 = 0$ and $w_j = \frac{1}{2}$ for $1 \leq j \leq k-1$. Then (2) reads

$$F_k\Big(\tfrac{1}{2} + it\Big) \;=\; \frac{\rho^k \sin(k\theta)}{2t},$$

so on the line F_k vanishes precisely where $\sin(k\theta) = 0$, i.e. at $\theta = n\pi/k$, i.e. at $t = \frac{1}{2}\tan(n\pi/k)$. The zeros therefore sit at

$$q \;=\; \tfrac{1}{2} + \tfrac{i}{2}\tan\!\left(\frac{n\pi}{k}\right), \qquad n = 1, \ldots, \left\lfloor \tfrac{k-1}{2} \right\rfloor,$$

and nowhere else on the line. The first three, at $k = 32$:

n	$t_n = \frac{1}{2}\tan(n\pi/32)$	$k\,t_n/\pi$
1	0.0492457016785821265	0.5016126
2	0.0994561836898290035	1.013052
3	0.151673341803671196	1.544932

[proved (Theorem 3.1(e)); the table is the formula evaluated, and **every entry is confirmed to be a zero of the direct sum to sixty digits** as an instrument control, `note1tab_r208`. The first entry is reproduced independently in three further experiments, `pinch_r202`, `trackm_r203`, `lambda_r206`.]

Two features of this list are used later. The spacing is uniform in θ — one rung per π/k — and $k\,t_n/\pi \to n/2$, so in t the rungs are asymptotically evenly spaced as well. Both facts fail as soon as the weights decay (Remark 4.5(i)), and the failure is not a small perturbation.

Corollary 3.3 (the pinch at $R = 1$, for eventually constant weights)**.** *[proved. Not immediate from Theorem 3.1: that theorem admits one exceptional weight w_0, whereas "eventually constant" admits finitely many. The two lines that close the gap are given.] Suppose $w_j = w > 0$ for all $j \geq j_0$, with $w_0, \ldots, w_{j_0-1} \geq 0$ arbitrary. Then $R = 1$, and F_k has a zero on $\operatorname{Re} q = \frac{1}{2}$ with $t_1 = \dfrac{\pi}{2k}(1 + O(1/k))$.*

Proof. Write $w_j = w + v_j$ for $j \geq 1$, so $v_j = 0$ once $j \geq j_0$, and put $B := 2\sum_{1 \leq j < j_0} |v_j|$. By Proposition 2.2,

$$F_k\Big(\tfrac{1}{2} + it\Big) \;=\; \underbrace{A + w\rho^k \frac{\sin(k\theta)}{t}}_{\text{Theorem 3.1}} + 2 \sum_{1 \leq j < j_0} v_j\, \rho^j \cos(j\theta),$$

and on $0 \leq \theta \leq 3\pi/2k$ we have $1 \leq \rho^j \leq \rho^{j_0} \to 1$ for fixed j_0, so the second sum is bounded by B in absolute value for all large k, *uniformly in k*. It therefore changes nothing that grows.

At $\theta = 3\pi/2k$ one has $\sin(k\theta) = -1$ and $t \leq (3\pi/4k)(1 + O(k^{-2}))$, so the main term is at most $A - 4kw/3\pi\,(1 + o(1))$, which beats $A + B$ once $k > 3\pi(A + B)/4w$: the value is negative there, and (c) of Theorem 3.1 survives with A replaced by $A + B$.

For the lower end, write $\theta = \pi/k - \varepsilon$ with $0 < \varepsilon < \pi/k$. Then $\sin(k\theta) = \sin(k\varepsilon) \geq \frac{2}{\pi}k\varepsilon$ and $t = \frac{1}{2}\tan\theta \leq \pi/2k\,(1 + O(k^{-2}))$, so the main term is at least $(4/\pi^2)wk^2\varepsilon\,(1 + o(1))$, which exceeds B as soon as $\varepsilon \geq \pi^2 B/(4wk^2)$. Hence $F_k > 0$ for $\theta \leq \pi/k - O(k^{-2})$, so $t_1 \geq \frac{1}{2}\tan(\pi/k - O(k^{-2})) = \frac{\pi}{2k}(1 - O(1/k))$. Combining, $t_1 = \frac{\pi}{2k}(1 + O(1/k))$. ☐ ☐

Remark 3.4 (what the two lines buy). [proved, being the corollary.] The broad statement is the one Conjecture 6.2's first branch needs to cite: "eventually constant" is a class, whereas Theorem 3.1 as stated is a family. The perturbation is bounded while the main term grows like k, which is why finitely many arbitrary weights cost nothing — *and it is worth noticing that the same estimate fails the moment infinitely many weights differ from w, which is exactly where §4 begins.*

4 Decaying weights: the second speed

Corollary 3.3 settles the constant-weight case. When the weights decay the pinch still occurs, and the rate is different.

Proposition 4.1 (the scale, derived)**.** *[derived, asymptotically and not to the standard of §3; **measured** at four exponents and five sizes with no fitted constant, `rate_r200`, `divide_r204`, `lambda_r206`.] Suppose $w_j \asymp (j+1)^{-s}$ with $s > 0$. In Proposition 2.2, $\rho^j = e^{2jt^2(1+O(t^2))}$, so the last term of the sum carries $w_k\rho^k \asymp k^{-s}e^{2kt^2}$. This is $o(1)$ below the scale $2kt^2 = s\log k$ and unbounded above it, so a zero cannot occur below that scale and the first one is expected at it:*

$$t_1 \;\asymp\; \sqrt{\frac{s\log k}{2k}}.$$

The scale is exact; the constant in front of it is not, and nothing here claims one.

The contrast with Corollary 3.3 is a factor $\asymp \sqrt{k \log k}$:

k	$t_1,\ w_j = \frac{1}{2}$	$t_1,\ w_j = (j+1)^{-1}$	$k\,t_1$ (left)	$k\,t_1$ (right)
64	0.0245634248847336271	0.1861996814519841	1.5720592	11.91677961
128	0.0122743110544627221	0.1409993892900241	1.5711118	18.04792183
256	0.00613623118978313762	0.106767212545108	1.5708752	27.33240641
512	0.00306800007881170098	0.07234505270561657	1.570816	37.04066699

The left column is $\frac{1}{2}\tan(\pi/k)$, which Theorem 3.1(e) gives exactly; the right is measured. The two right-hand columns are the diagnostic: $k\,t_1$ *converges* to $\pi/2 = 1.5707963$ on the left and *grows* on the right. That divergence is the whole content of the two-regime statement — the second family does not merely pinch more slowly, it pinches at a different power of k. [left column proved, `trackm_r203`; right column measured, `divide_r204`.]

4.1 Where the dividing line is

Proposition 4.2 (the line is decay, not summability). *[the refutation is* ***measured*** *and decisive,* `divide_r204`*; the positive statement is* ***conjectured****, see Conjecture 6.2.] The dividing line between the two rates is whether $w_j \to 0$, and* not *whether $\sum_j w_j$ converges. The harmonic profile $w_j = (j+1)^{-1}$ has $\sum_j w_j = \infty$ and takes the $\sqrt{\ }$ rate: $k\,t_1$ measures* 11.9, 18.0, 27.3, 37.0, 55.6 *at $k = 64, \ldots, 1024$, growing rather than approaching $\pi/2 =$* 1.5707963*. The same holds for $w_j = \big((j+2)\log(j+2)\big)^{-1}$, which diverges more slowly still ($k\,t_1$ reaching* 62.2*).*

Remark 4.3 (how the wrong line survived, and what the mechanism said). [methodological; no mathematical content beyond Proposition 4.2.] Every divergent-side profile we had computed had *constant* weights and every convergent-side one decayed like a power, so the data were equally consistent with a different statement — *constant versus decaying* — that had not been written down. The two separate only on profiles that decay without summing, and none had been computed. *A dividing line is a claim about the profiles that lie on it, and evidence from far to either side does not test it.*

Moreover the estimate in Proposition 4.1 uses w_k and never uses $\sum_j w_j$: summability was never one of its hypotheses. *When a conjecture and its own mechanism disagree about what the hypothesis is, the mechanism is the one that was derived.*

4.2 The constant at $s = 1$, unsettled

Proposition 4.1 names the scale but not the constant in it. Writing

$$\lambda_{\mathrm{eff}}(k,s) \;:=\; \frac{2k\,t_1^2}{\log k},$$

the derived prediction is $\lambda_{\mathrm{eff}} \to s$. We report the state of that question rather than a conclusion.

candidate	evidence	verdic
$\lambda_{\mathrm{eff}} \to s - \frac{1}{2}$	$\lvert\lambda_{\mathrm{eff}} - (s - \frac{1}{2})\rvert$ stays in $[0.34, 0.48]$ and barely shrinks	**not r**
$\lambda_{\mathrm{eff}} \to s$	$\lvert\lambda_{\mathrm{eff}} - s\rvert$ reaches 0.16, 0.14, 0.099, 0.093, 0.018 at $s = 1, \frac{3}{2}, 2, 3, 4$	consis
correction of size $1/\log k$	$(\lambda_{\mathrm{eff}} - s)\log k$ does not settle	**not r**
correction of size $\log\log k/\log k$	nor does $(\lambda_{\mathrm{eff}} - s)\log k/\log\log k$	**not r**

[measured over $k \le 2048$ at five exponents, `lambda_r206`; the decision rule and the two candidate shapes were registered before the run.]

The measurements behind that table, in full, are:

k	λ_{eff}, $s=1$	λ_{eff}, $s=2$	λ_{eff}, $s=4$
64	1.0670656	2.8134049	6.3869596
128	1.0489407	2.0877026	5.3173568
256	1.0525199	2.0014877	4.3314313
512	0.85911175	1.978608	4.3134221
1024	0.87254853	1.86631	4.071501
2048	0.83659266	1.9008311	3.9822736

[measured, `lambda_r206`; 40 working digits; the first zero located by scanning upward from $t=0$.]

Each column is near its target and none is monotone, and the departures at the largest k are of the same size as those at $k=128$. *We report the sequences rather than a fitted constant*: a fit over these would produce a number the data do not support, and the honest content is that the scale is right and the constant is not established. Remark 7.2 explains *why* these columns sit near s while the mechanism's own asymptote is $s-\frac{1}{2}$, and in doing so withdraws our refutation of that asymptote.

Remark 4.4 (on the observable). [methodological.] The natural thing to print is t_1 divided by $\sqrt{s \log k/2k}$, and we did so for four rounds. That ratio is $\sqrt{\lambda_{\text{eff}}/s}$, and **a square root halves relative error**: a 14% departure in the quantity the derivation names appears as 7% on the page. *Report the quantity the mechanism names; a monotone transform is not a neutral change of units but a lens with a computable magnification.*

Remark 4.5 (two readings we can rule out). [measured, `envelope_r206b`.] Two natural explanations for the non-monotonicity fail.

(i) *Quantisation.* One might expect t_1 to be pinned to a ladder of troughs spaced π/k, as in Theorem 3.1(e), giving jitter $\pi/(kt_1)$. But when the weights decay the zeros are *not* evenly spaced: the measured gaps run 0.17 to 0.73 in units of π/k, at $s=1$ and $s=2$ alike. **The even ladder is a constant-weight phenomenon.**

(ii) *An envelope threshold.* One might locate the first zero as the first t with $2|G_k(1+2it)|=1$. No such t exists: $2|G_k|-1$ equals $+11.2$ at $t=0$ for $s=1$, $k=1024$, since $|G_k(1)|$ is half the value at the symmetry point. The zero is produced by the *phase* turning, not by the envelope growing.

5 Two theorems, and the shape of the answer

Since the first version of this note, the decaying case has stopped being a table of measurements. Two statements are now proved, and together they fix the shape of the answer.

Theorem 5.1 (every non-increasing profile)**.** *[proved. Two independent derivations along disjoint routes; the identity it rests on was verified against direct summation to sixty digits, including at the zeros.] Let $w_0 \ge w_1 \ge \cdots \ge w_{k-1} \ge 0$. If $\arctan(2t) \le \pi/k$ then $F_k(\frac{1}{2}+it) \ge 1$. Consequently the first zero satisfies $t_1 > \frac{1}{2}\tan(\pi/k)$.*

The proof is two lines from an exact identity obtained by summation by parts: writing $D_j = w_{j-1} - w_j$ and $z = 1+2it = \rho e^{i\theta}$,

$$F_k(\tfrac{1}{2}+it) \;=\; 1 + \frac{1}{t}\Big[\, w_{k-1}\rho^k \sin(k\theta) + \sum_{j=1}^{k-1} D_j\, \rho^j \sin(j\theta) \Big]. \tag{3}$$

If the profile is non-increasing every D_j is non-negative, and on $\theta \le \pi/k$ every $j\theta$ with $j \le k$ lies in $(0, \pi]$, so every sine is non-negative and the bracket cannot be negative. Theorem 3.1 is the special case in which the sine series is empty. *Monotonicity is doing work and is not necessary*: an increasing ramp drives F_k well below 1, while a bump and an alternating profile stay above it without being monotone.

Theorem 5.2 (the second speed, for power weights)**.** *[proved. Written proof and an independent re-derivation along a different route; the two moment identities it uses are additionally replayed through the Lean kernel.] Fix $s > 1$ and let $w_j = (j+1)^{-s}$. Then*

$$\frac{2k\, t_1(k)^2}{\log k} \longrightarrow s - \tfrac{1}{2} \qquad (k \to \infty).$$

The mechanism is visible in (3). The coefficients are the weight *decrements*, which decay one power faster than the weights themselves, so the sine series concentrates near $j \asymp 1/2t$ rather than near $j \asymp k$; its limit is $2\zeta(s)$, and the top term must supply the rest. The tail is controlled by cancellation and not by absolute values — the sum of $|D_j \rho^j|$ over the tail diverges for $1 < s < 2$ at this scale — which is possible because $D_j \rho^j$ has exactly one interior minimum, D_j being log-convex as the integral of a log-convex function over a unit interval.

Remark 5.3 (what the two speeds are, and where they meet). [$s > 1$ **proved** (Theorem 5.2); $s < 1$ **measured**, decisively, the two candidate limits differing by more than twenty times the observable's own resolution; $s = 1$ **derived** by two independent routes.] For $w_j = (j+1)^{-s}$ the effective exponent $\lambda = 2kt_1^2/\log k$ appears to satisfy

$$\lambda_\infty(s) \;=\; \max\Big(\frac{s}{2},\; s - \frac{1}{2}\Big),$$

with the kink at $s = 1$ — exactly where $\sum_j w_j$ stops converging. Above the kink the convergent weight sum sets the level and the top term alone places the zero; below it the sum diverges and drags the zero outward. The constant governing $s < 1$ is $2^{s-1}\pi/(\Gamma(s)\cos(\pi s/2))$, whose surviving poles sit at the odd integers.

6 The general question

Problem 6.1 (does $R = 1$ force the pinch?)**.** [**proved** for constant weights (Corollary 3.3); **measured** for $w_j \asymp (j+1)^{-s}$ at four exponents; **open** in general.] Let $w_j \ge 0$ with $R = 1$. Do the zeros of F_k approach $q = \frac{1}{2}$ as $k \to \infty$?

Conjecture 6.2 (two speeds)**.** [conjectured; the constant-weight half is Corollary 3.3 and is proved, the decaying half is derived and measured, and the constant at $s = 1$ is unsettled (§4.2).] Let $w_j \ge 0$ with $R = 1$. Then F_k has a zero on $\operatorname{Re} q = \frac{1}{2}$ approaching $q = \frac{1}{2}$, at rate

$$t_1 \sim \frac{\pi}{2k} \quad \text{if } w_j \not\to 0, \qquad\qquad t_1 \sim \sqrt{\frac{s \log k}{2k}} \quad \text{if } w_j \asymp (j+1)^{-s},\ s > 0.$$

The two displayed classes are illustrative and do not exhaust $R = 1$. They omit, for instance, profiles decaying more slowly than every power, such as $w_j = 1/\log(j+2)$. We expect such profiles on the second branch with a local exponent in the role of s; that expectation is measured only at $w_j = ((j+2)\log(j+2))^{-1}$ and is deliberately **not** part of the conjecture.

The missing analytic ingredient, named rather than left to be located. What is absent for the general case is the non-vanishing of the limit of G on its circle of convergence. Hurwitz's theorem gives non-accumulation on compact subsets of the region of analyticity; accumulation *on* the boundary needs that non-vanishing, and it is a statement about G alone.

A fingerprint of the sum, with its scope. Sections of a single geometric series have zeros at angular spacing $2\pi/k$; in the constant-weight case ours sit at $\theta_n = n\pi/k$, twice as densely. That doubling is what $\sin(k\theta)$ does and $e^{ik\theta}$ does not: taking a real part folds $e^{ik\theta}$ into $\sin(k\theta)$, whose zeros are twice as dense. **So the doubled ladder is exactly the part of the phenomenon the single-section theory does not describe** — and, by Remark 4.5(i), it is a statement about profiles with $w_j \not\to 0$ and not a general one. [proved for constant weights (Theorem 3.1(e)); the scope restriction is measured, `envelope_r206b`.]

7 What a proof of the decaying case would have to supply

Theorem 3.1 works because the sum is geometric and collapses. When the weights decay there is no such collapse, and we record what we tried, so that the next attempt does not repeat it.

The naive balance is wrong. Near the first zero the sum in Proposition 2.2 is dominated by j close to k, where the phases $j\theta$ turn rapidly. Integrating by parts once suggests the oscillatory part has size $\asymp w_k\rho^k/\theta$ rather than $w_k\rho^k$, and balancing *that* against the constant term gives $k^{-s}e^{2kt^2} \asymp t$, hence $\lambda_{\text{eff}} \to s - \frac{1}{2}$.

Proposition 7.1 (the naive balance, and why it was wrongly refuted)**.** ***[withdrawn as a refutation.*** *It was recorded as one on the grounds that $|\lambda_{\text{eff}} - (s-\frac{1}{2})|$ stays in $[0.34, 0.48]$ over $k \le 2048$ without shrinking. Remark 7.2 shows that band is exactly what the balance predicts at those k. `lambda_r206`, `rung0_r211`.] The balance $k^{-s}e^{2kt^2} \asymp t$ gives $\lambda_{\text{eff}} \to s - \frac{1}{2}$ with an approach of size $(\log\log k)/(2\log k)$, which at $k = 1024$ is 0.13966. Over the computable range it therefore places λ_{eff} near s, not near $s - \frac{1}{2}$, and the measurement cannot separate the two.*

Remark 7.2 (the two candidate shapes are one model at different k). [the model is **derived**; its agreement with the measurement is **measured** at 25 pairs (k, s) with no fitted parameter, worst error 9.9%, `rung0_r211`. The asymptote is a consequence of the model and is **conjectured**.] Split G_k into a head and a tail. For small j the terms contribute $H := \sum_{j<k} w_j$, real and positive; near $j = k$ the weights are nearly constant over the range on which $|z|^j$ changes, so the tail is geometric,

$$\sum_{j \text{ near } k} w_j z^j \;\approx\; w_{k-1}\frac{z^k}{z-1}, \qquad \text{modulus } T := \frac{w_{k-1}\rho^k}{2t}, \qquad \text{argument } k\theta - \frac{\pi}{2},$$

the quarter turn coming from $\arg(z-1)^{-1} = -\pi/2$ exactly. Hence $\operatorname{Re} G_k \approx H + T\sin(k\theta)$, and a zero of F_k first becomes possible when $T = H + \frac{1}{2}$. **Solving that for t, with nothing fitted, reproduces the measured first zero to within 9.9% at every (k, s) tested.**

Asymptotically the same equation reads $k^{\lambda - s} = 2t(H + \frac{1}{2})$, so

$$\lambda_{\text{eff}} - s \;=\; -\tfrac{1}{2} + \frac{\log\log k}{2\log k} + O\!\left(\frac{1}{\log k}\right).$$

The two shapes registered in §4.2 are therefore the same statement at different k: the correction term is 0.13966 at $k = 1024$ and decays like $1/\log k$, so over any computable range λ_{eff} sits near s while tending to $s - \frac{1}{2}$. *Our own refutation of the second shape tested a limit that this model says is approached far too slowly to be seen, and is withdrawn.*

Two instruments that do not exist or do not apply. Remark 4.5 records both: the zeros are not a uniform ladder once the weights decay, so no quantisation argument is available; and the first zero is not an envelope crossing, because $2|G_k(1+2it)|$ already exceeds 1 at $t=0$ by a wide margin. **The zero is a phase event**: at the first zero, $\operatorname{Re} G_k = -\frac{1}{2}$ while $|G_k|$ remains large, so $\cos(\arg G_k) = -1/(2|G_k|)$, which is close to 0 from below. A treatment in terms of $\arg G_k(1+2it)$ therefore has the right shape — but a linear approximation to that phase predicts a crossing at $t \asymp k^{-(2-s)}$, which is not the observed scale, so any such treatment owes an account of how the amplitude redistributes with t. [derived, at the level of a sketch; the linear-phase estimate is recorded because it is wrong in a specific and informative way.]

8 What is verified, and how

This note carries a status at every statement, and the statuses are meant literally. Their evidence is as follows.

status	what stands behind it here
proved	a written proof, plus an independent re-derivation along a disjoint route by a second re
derived	an asymptotic argument, stated as such; not to the standard of the proofs in §3
measured	a computation with a committed log, with the range stated; a range is not a limit
refuted	a pre-registered falsifier fired
conjectured, *open*	neither of the above

Two consequences of taking this seriously are visible in the text above. Theorem 3.1 says *proved* because it was written by one reader and re-derived from $G_k(z) = w_0 + w(z-z^k)/(1-z)$ by another who did not read the first derivation; Proposition 4.1 says *derived* and not *proved*, although we believe it, because an asymptotic balance is not a proof.

Remark 8.1 (the record of this note's own errors). [methodological; no mathematical content.] Three statements that appear above in corrected form were previously published, under a DOI, in incorrect form: an oscillator given as cos where the geometric sum gives sin (the phase of $1/(z-1)$ is exactly $-\pi/2$, and a quarter turn moves a ladder by half a rung); an equivalence "$R \geq 1$ if and only if $\sum_j w_j < \infty$", which is false exactly at $R=1$ where both cases occur; and the dividing line of Conjecture 6.2 given as summability rather than decay (Remark 4.3). Each was corrected at its statement in the version where it appeared, and named in that version's release notes; the record is public in the repository identified in the disclosure below.

We record this because a note whose apparatus is part of its claim should say what that apparatus has caught. Two of the three were found not by a check but by building something new and noticing that it contradicted a sentence written elsewhere.

9 Three questions

1. **Does a sum of two sections inherit boundary accumulation?** Precisely: for $w_j \geq 0$ with $R=1$, does F_k have zeros approaching $q=\frac{1}{2}$? (Problem 6.1. Answered for $w_j \not\to 0$; measured for $w_j \asymp (j+1)^{-s}$; open in general.)

2. **Is the dividing line exactly $w_j \to 0$?** We have refuted summability as the criterion and verified the decay criterion on both sides, including on profiles that decay without

summing — but "$w_j \to 0$" has not been shown to be *sharp*. A profile with $w_j \to 0$ taking the $\pi/2k$ rate would refute Conjecture 6.2.

3. **What is the constant at $s = 1$?** The scale $\sqrt{s \log k/2k}$ is derived and measured at four exponents; at $s = 1$ the constant is not established (§4.2). *This is the cheapest of the three to attack and the one we most expect to be closed by someone else.*

10 Where we looked in the literature

So that the claim "we found nothing covering sums of sections" can be checked or refuted cheaply, here is its extent. We consulted the classical statements of Jentzsch (1914) and Szegő, and surveyed the extensions of Jentzsch–Szegő to Dirichlet series, Kapteyn series, Neumann series, sections along analytic curves, and the ultrametric setting. All of these concern the zeros of the sections of *one* series, or of one series composed with a map. We did not find a treatment of

$$\alpha\, G_k(\varphi_1(q)) + \beta\, G_k(\varphi_2(q)) + \gamma$$

for affine φ_1, φ_2 — in particular, of the conjugate pair that (1) produces on a line. [a negative result about our search, not about the literature; stated so that it can be overturned by one reference.]

11 Numerical conventions

Every number in this note is copied from a committed log, and every script named in a status is in the repository with its log beside it. Computations use `mpmath` at 30–120 decimal digits, with the working precision printed; where a quantity is compared across precisions, the number of agreeing digits is printed rather than asserted. Zeros on the line are located by scanning t upward *from zero* and bisecting the first sign change, so that "first" is a claim about $(0, t_{\max})$ and not about a window; where a claim concerns zeros off the line as well, it is certified by an argument-principle count on a circle about $q = \frac{1}{2}$.

Each experiment registers its falsifiers before it runs, and at least one of them tests the *instrument* against a value proved elsewhere in this note — typically Theorem 3.1(e), whose zero set is exact. In the run that produced §4 the three falsifiers testing the law all passed while both instrument controls failed, on an off-by-one in the weight index; the law-testing falsifiers alone would have confirmed a wrong apparatus.

Tool and computational resource disclosure

This note was produced by one person using AI models as tools under his direction. Draft text, proofs, code and numerical experiments were generated in collaboration with large language models (Anthropic's Claude, models `opus-5` and `fable-5`); the author directed the work, chose what to keep, and is responsible for the content.

Because work produced this way cannot be trusted on the author's word, the verification apparatus is part of the artefact and is public: every experiment writes a committed log, every number in this note is copied from one, statuses are carried at each statement, and a failure ledger recording the project's own errors — including three corrections to statements that had already been published under a DOI — is in the repository. Theorem 3.1 was written by one model and independently re-derived along a disjoint route by the other before being labelled *proved*.

Everything named in a status above — every script, its log, the ledger, and the twenty-one mechanical checks that gate each commit — is at https://github.com/mathlab0911/arithmetic-landscapes. *A reader who wants to disbelieve a number in this note should be able to find the line that produced it without asking the author.*